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“Jnana Sangama”, Belgaum-590014, KARNATAKA, INDIA



Project Phase I Report - BCS685

On

***Human Movement Functional Assessment
and Analysis System using Wearable IoT
Sensors and AI***

***A project report submitted in partial fulfillment of the requirements for the
award of the degree of Bachelor of Engineering in Computer Science and
Engineering of Visvesvaraya Technological University, Belgaum.***

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ABSTRACT

In this project, we are developing an intelligent system that helps clinicians and professional organizations automatically quantify and analyze human movement using wearable IoT sensors and AI. The core idea is to analyze biomechanical orientation at a high-frequency level to identify joint angles, stability, and movement speed, enabling better monitoring of physical readiness and rehabilitation progress. Professional fields such as military selection and clinical orthopedics often face challenges in tracking recovery accurately, ensuring rehabilitation goals are met, and identifying subtle movement anomalies across varied demographics. Our solution addresses these issues by utilizing a wearable ESP32-integrated MPU6050 node, which serves as a high-precision motion capture reference. When new movement data is analyzed, the system compares it to the user's historical records and standardized "Ideal Person" benchmarks based on age, gender, and physical characteristics. This enables organizations to verify if a candidate is physically fit for duty or if a patient's recovery trajectory is optimal. Additionally, the system generates a comprehensive analysis report to track longitudinal trends in stability and range of motion. This makes the process of functional assessment fast, objective, and scalable, eliminating the need for subjective manual review. The goal of the project is to create a practical, high-precision platform that enhances how institutions monitor physical movement, safeguard musculoskeletal health, and improve the overall effectiveness of rehabilitation and selection strategies.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Human movement quality is a critical indicator of musculoskeletal health, functional mobility, and overall physical well-being. To address the limitations of traditional, subjective assessment methods, this project introduces the development of a wearable, AI-powered system designed for objective, real-time functional movement analysis. The device utilizes advanced Inertial Measurement Units (IMUs)—specifically the MPU6050—paired with a high-performance ESP32 microcontroller. These components work in tandem to capture precise orientation, acceleration, and angular velocity data, allowing for the continuous monitoring of limb dynamics and postural changes. The core of the system is an AI-driven analytical engine that processes raw sensor data to extract meaningful biomechanical metrics, such as joint angles, Range of Motion (ROM), and movement smoothness. Unlike traditional clinical tools like manual goniometers, the AI model compares user data against reference functional mobility profiles to detect anomalies like lack of symmetry or compensation behaviors. This data-driven approach improves the accuracy of assessments and provides a quantitative basis for evaluating physical performance and recovery. To ensure user comfort and clinical utility, the wearable is designed to be lightweight, non-intrusive, and suitable for long-term daily or rehabilitative use. The system supports wireless communication via Wi-Fi to transmit data to a real-time dashboard, facilitating immediate feedback for both users and physiotherapists. By optimizing the analysis for edge computing and cloud integration, the platform ensures low-latency processing and longitudinal tracking of movement trends. Future iterations aim to expand the system's capabilities into sports science and preventive healthcare, transforming it into a comprehensive tool for autonomous biomechanical monitoring.

1.2 Objectives

The primary aim of this project is to implement a real-time, intelligent, and objective functional movement assessment system using wearable IoT sensors and AI models. While

the current phase successfully implements the hardware-to-software movement assessment framework, the project is designed to integrate advanced Machine Learning (ML) models to provide significantly better analysis of biomechanical data. The system utilizes ESP32-based sensor processing for immediate assessment and is transitioning toward an AI-driven analytical layer for pattern recognition and anomaly detection. The specific objectives are as follows:

- **Establish a Wearable IoT Framework (Currently Implemented):**

To utilize MPU6050 Inertial Measurement Units (IMUs) and the ESP32 microcontroller to capture high-precision acceleration and angular velocity data, forming the hardware foundation for movement tracking.

- **Compute Real-Time Biomechanical Parameters (Currently Implemented):**

To implement processing algorithms that calculate joint angles and Range of Motion (ROM), providing an immediate, objective quantification of physical movement dynamics.

- **Integrate Machine Learning Models for Better Analysis:**

To deploy ML models that move beyond raw data capture to provide deeper insights into movement quality. By training on diverse datasets, these models will identify subtle gait anomalies and movement inefficiencies that are difficult to detect via manual assessment.

- **Automated Anomaly Detection and Functional Comparison:**

To develop an AI-based analytical engine that compares user performance against reference mobility profiles. This enables the system to provide "better analysis" through automated risk scoring, detection of compensation behaviors, and symmetry evaluation.

- **Develop a Comprehensive Analytics and Feedback Dashboard:**

To create a modern web-based interface that visualizes real-time telemetry and maintains historical records. This allows for longitudinal tracking of recovery and the visualization of AI-generated insights for both users and health professionals.

1.3 Purpose, Scope, and Applicability

1.3.1 Purpose

The purpose of this project is to develop a wearable, AI-powered functional movement assessment system designed to objectively quantify and analyze human mobility. As movement quality is a critical indicator of musculoskeletal health, there is a significant need for real-time, intelligent systems that move beyond subjective manual observations to provide data-driven insights for rehabilitation and performance monitoring.

Traditional assessment methods often rely on manual tools like goniometers, which are clinic-dependent and prone to human error. This project addresses these challenges by utilizing embedded IoT sensors and AI algorithms for continuous, automated movement analysis in real time. The system is designed to provide "better analysis" by identifying subtle biomechanical deviations that are often missed during visual inspections.

The system captures high-precision data from wearable devices to compute joint angles and Range of Motion (ROM). By integrating an AI analytical layer, the platform can eventually compare user movements against reference functional profiles, allowing for the detection of compensation behaviors and providing actionable feedback to both users and healthcare professionals to ensure effective physical recovery.

1.3.2 Scope

This project focuses on building a real-time biomechanical analysis ecosystem combining sensor hardware, AI-based processing, and web-based connectivity. It includes the development and integration of the following components:

- **Wearable IoT Device** equipped with the ESP32 microcontroller and MPU6050 IMU sensors for orientation tracking.
- **Machine Learning Models** (Upcoming integration) for advanced pattern recognition, anomaly detection, and enhanced movement classification.
- **Node.js and Express Backend** to manage high-frequency data ingestion, project logic, and system communication.
- **Supabase (PostgreSQL)** for secure storage of sensor telemetry, user profiles, and historical assessment records.
- **React Web Dashboard** featuring Three.js for 3D motion visualization and real-time telemetry charts.

The system supports the following key operations:

- Continuous acquisition of acceleration and angular velocity data from the wearable sensor.
- Real-time computation of joint angles, Pitch, Roll, and Range of Motion (ROM).
- Generation of movement quality scores based on smoothness, symmetry, and stability.
- Management of historical test results for longitudinal tracking of rehabilitation progress.

The current version focuses on the successful implementation of the movement assessment framework and ROM tracking. Future work involves deploying specialized ML models to provide deeper diagnostic analysis and automated mobility risk scoring.

1.3.3 Applicability

This system is highly applicable across various domains where physical performance and functional mobility are paramount:

- **Physiotherapy and Rehabilitation:** To monitor patient progress objectively and adjust recovery plans based on precise ROM data.
- **Sports Science and Athletics:** For tracking athlete form, identifying movement inefficiencies, and preventing overstrain injuries.
- **Elderly Care and Wellness:** To identify early signs of mobility decline or gait instability, facilitating preventive interventions.
- **Orthopedic Clinical Assessment:** As a low-cost, digital alternative to expensive motion-capture labs for routine joint evaluations.
- **Home-Based Tele-Rehabilitation:** Enabling patients to perform exercises correctly at home while providing remote data to their clinicians.
- **Occupational Health:** For assessing ergonomic risks and movement patterns in industrial or corporate environments to prevent workplace injuries.

The system is modular and extensible, with potential for future enhancements such as multi-sensor body networking, AI-based gait prediction, and integration with advanced health monitoring sensors.

1.4 Organization of the Report

This report is structured in a clear and logical manner to guide the reader through the background, motivation, system design, and evaluation of the Human Movement Functional

Assessment and Analysis System. Each chapter builds on the previous one to provide a comprehensive understanding of the project and how the integration of wearable IoT and upcoming AI models contributes to the objective quantification of biomechanics.

- i. **Chapter 1: Introduction** - Provides an overview of the project, highlighting the motivation behind developing an objective, sensor-based system for functional movement assessment. This chapter outlines the purpose, scope, and applicability of the system, introducing the transition from raw data capture to data-driven insights.
- ii. **Chapter 2: Problem Statement** - Defines the central challenge of manual and subjective mobility assessments in clinical settings. It highlights the limitations of traditional tools like goniometers, the risks of undetected movement compensations, and the need for a continuous, real-time monitoring solution.
- iii. **Chapter 3: Review of Existing Solutions** - Surveys previous approaches to movement tracking, including clinic-based motion capture labs and basic wearable pedometers. It discusses their shortcomings in terms of cost, accessibility, and the inability to provide deep biomechanical analysis such as movement smoothness and symmetry.
- iv. **Chapter 4: Proposed Solution** - Introduces the proposed framework, leveraging wearable IMU sensors and a future AI analytical layer. It explains the system's focus on three core assessments—Left Arm Flexion, Right Arm Extension, and Torso Rotation—and how the upcoming integration of ML models will provide "better analysis" of movement anomalies.
- v. **Chapter 5: Methodology** - Details the technical methodology followed, including sensor data acquisition using acceleration and angular velocity, the mathematical computation of joint angles and Range of Motion (ROM), and the logic for the upcoming AI-based functional comparison pipeline.
- vi. **Chapter 6: Implementation Details** - Describes the implementation of the assessment ecosystem. This includes the wearable sensor setup using the ESP32 and MPU6050, the Node.js backend for high-frequency data ingestion, Supabase for cloud storage, and the React-based dashboard for real-time telemetry visualization through dynamic charts.
- vii. **Chapter 7: Results and Discussion** - Presents the results obtained during the cur-

rent assessment phase. It includes ROM accuracy benchmarks, real-time telemetry chart performance, and data analysis of the three standardized movement tests. The discussion addresses the current implementation strengths and the specific benefits expected from the next-phase ML integration.

viii. Chapter 8: Conclusion and Future Scope - Summarizes the major contributions of the project, specifically its impact on home-based rehabilitation and objective mobility scoring. Future scope highlights the deployment of advanced ML models for predictive injury analysis and holistic health monitoring.

ix. Bibliography - Lists all research papers, documentation, and technical frameworks referred to throughout the project, ensuring academic integrity and grounding the project in established biomechanical research.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Quantifying the quality of human movement is a fundamental requirement in clinical rehabilitation, sports science, and physical therapy. Movement quality serves as a primary indicator of musculoskeletal health, functional mobility, and recovery progress. Traditionally, the assessment of joint flexibility and Range of Motion (ROM) has relied on manual tools such as goniometers. However, these methods are often subjective, clinic-dependent, and lack the precision required for continuous or longitudinal monitoring. Consequently, there is a growing demand for objective, sensor-based systems that can provide real-time biomechanical insights.

The integration of the Internet of Things (IoT) and Artificial Intelligence (AI) has revolutionized how movement is tracked and analyzed. Compact wearable devices, equipped with Inertial Measurement Units (IMUs) containing accelerometers and gyroscopes, can capture human motion with high precision. When these data streams are processed using microcontrollers like the ESP32, it becomes possible to compute orientation and joint angles in real time. The ultimate goal of such systems is to move beyond simple data logging toward sophisticated AI-driven analysis that can identify anomalies, such as movement compensation or asymmetry, which are often invisible to the naked eye.

Recent research has extensively explored the use of IMU sensors for joint kinematics. Studies have demonstrated that tri-axial sensor data, when combined with filtering algorithms, can achieve high accuracy in estimating limb orientation in controlled environments. This validates the use of wearable sensors for standardized mobility tests, including arm flexion, extension, and torso rotation. However, the transition from raw data acquisition to "better analysis" requires robust backend systems and advanced machine learning models to handle the variability of human movement.

The development of modern biomechanical systems often utilizes frameworks like Node.js for high-frequency data ingestion and cloud-connected databases such as Supabase for secure storage. These technologies enable the creation of interactive dashboards that provide users and healthcare professionals with real-time telemetry and historical trend vi-

sualization. By moving away from manual review and toward automated data processing, these platforms enhance the transparency and effectiveness of physical therapy and athletic training.

Machine learning plays a pivotal role in elevating the utility of movement assessment systems. By training models on diverse datasets of functional mobility, researchers have been able to differentiate between healthy movement patterns and dysfunctional ones. Advanced models like Random Forests or Recurrent Neural Networks are increasingly being used to provide predictive insights into recovery timelines and injury risks. These models allow for an objective scoring of movement quality based on metrics such as smoothness, stability, and Range of Motion.

Despite significant advancements, several technical challenges remain. Most wearable systems must balance the trade-off between battery life and the high sampling rates required for accurate motion capture. Furthermore, sensor drift and environmental interference can impact the precision of joint angle calculations over time. There is also a continuous need for lightweight, non-intrusive hardware that users can wear comfortably during rehabilitative exercises.

This chapter sets the context for developing a comprehensive Human Movement Functional Assessment and Analysis System. By reviewing the evolution from manual tools to IoT wearables and identifying the current role of AI in biomechanics, this report lays the foundation for an optimized, data-driven solution. The focus remains on bridging the gap between hardware-based data capture and sophisticated machine learning analysis to provide a holistic approach to human functional evaluation.

2.2 Objective of Literature Survey

The primary objectives of conducting the literature survey for the Human Movement Functional Assessment and Analysis System are as follows:

- i. To study current techniques and technologies used in wearable health monitoring systems, particularly those aimed at tracking human activity and capturing joint kinematics using sensor data such as accelerometers and gyroscopes.
- ii. To analyze the effectiveness of various machine learning and deep learning models in accurately estimating human joint kinematics and classifying complex movement patterns for standardized physical tests [4].

- iii. To identify limitations in existing systems, such as the high cost of clinical motion-capture labs, lack of portability, and the difficulty of maintaining accuracy during real-world movement assessments.
- iv. To examine the role of real-time embedded processing and filtering algorithms in improving the accuracy of joint angle estimation and the energy efficiency of wearable devices [1].
- v. To gather insights into the detection of movement anomalies and irregularities through advanced analytical models to move toward "better analysis" of functional mobility [3].

2.3 Summary of Literature Survey

The literature survey on wearable systems for human movement assessment reveals significant advancements in sensor-based monitoring and AI-driven biomechanical analysis. The key focus areas of the reviewed research include high-precision sensor selection, machine learning models for pattern recognition, real-time data processing at the edge, and integration with modern IoT platforms. Each study contributes uniquely to addressing the challenges of objective movement quantification, focusing on joint angles, Range of Motion (ROM), and the transition from raw telemetry to sophisticated analytical insights.

[1] IoT-Enabled Movement Assessment Using LSTM and Edge Computing

AUTHORS: Pravin Kulurkar et al., 2023

Feature Focus: LSTM-Based Edge Intelligence for Real-Time Biomechanics

This study introduces an IoT-based movement monitoring system using wearable accelerometer sensors combined with edge computing and Long Short-Term Memory (LSTM) models for efficient analysis. The authors propose a low-power wearable node using the LSM6DS0 MEMS sensor that captures 3D motion data. The LSTM model was trained to identify movement sequences, achieving high accuracy in classifying functional tasks. A major innovation is offloading computational tasks from wearables to gateways, enabling reduced energy usage while maintaining high-quality data for "better analysis." This architecture supports real-time telemetry to dashboards while ensuring sensor longevity and usability.

[2] Comprehensive Survey of IoT-Based Motion Tracking Systems

AUTHORS: Mohamed Esmail Karar, Hazem I. Shehata, Omar Reyad, 2022

Feature Focus: Taxonomy of Sensors and Algorithms in Movement Analysis

This extensive survey provides a classification of movement assessment systems based on sensor types, ML techniques, and IoT integration. The authors evaluate popular sensors like accelerometers and gyroscopes for capturing limb dynamics. They highlight machine learning models including ANN and SVM for both posture prediction and movement classification. A key contribution is the identification of dataset limitations and the lack of standardization in evaluating ROM. The authors propose using lightweight models with edge computing to overcome latency in wearable devices.

[3] Visual AI for Movement Assessment and Posture Recognition

AUTHORS: Hui Jin, Weijun Zhou, Yuh-Chung Lin, 2024

This paper presents a visual assessment system designed to improve movement monitoring through 3D spatial awareness and real-time object detection. The proposed solution combines spatial mapping with CNN-based models to excel in identifying movement patterns through visual recognition. The system is implemented as a mobile application to perform on-device inference. A key component involves the training of models on large-scale posture datasets to distinguish between normal and dysfunctional movement patterns. This level of accuracy is critical for applications where visual feedback adds important context to sensor data.

[4] Accelerometer-Based Movement Assessment Using Edge AI Architecture

AUTHORS: Osama Zaid Salah, Sathish Kumar Selvaperumal, Raed Abdulla (2022)

This paper investigates an innovative wearable system that leverages Edge AI and low-power networks to deliver efficient, low-latency health monitoring. The authors propose a system where movement analysis occurs locally on the wearable device, utilizing microcontrollers integrated with 9-axis IMUs. A significant innovation is the deployment of a lightweight CNN model directly onto resource-constrained hardware to provide real-time assessment. This architectural approach minimizes latency and dependency on cloud networks while extending the operational lifespan of the wearable through optimized power consumption.

[5] Evolution of Human Movement Detection Systems

AUTHORS: Xueyi Wang, Joshua Ellul, George Azzopardi (2020)

This survey paper reviews the state-of-the-art in human movement detection systems, focusing on wearable sensor-based and vision-based approaches. The paper identifies key challenges such as high false alarm rates and the difficulty of balancing accuracy with real-time responsiveness. Among the pioneering works discussed are systems using accelerometers placed on the trunk to extract depth information and movement patterns. The authors emphasize the role of sensor fusion to improve robustness and call for a balanced approach combining accuracy, privacy, and user acceptance in rehabilitative technologies.

[6] Ensemble Deep Neural Network Approach for Movement Classification

AUTHORS: Zahir Mohammad, Arif Reza Anwary, Muhammad Firoz Mridha, et al. (2023)

This paper proposes an advanced wearable system using an ensemble deep neural network that combines CNNs and RNNs. The system is designed to anticipate movement phases—allowing for preventive clinical actions. This is achieved through a specialized ensemble architecture trained to recognize specific movement states. The researchers utilize large datasets to enhance model generalization. The ensemble design reduces overfitting and computational complexity, which is critical for real-world deployment where “better analysis” of biomechanical states is required.

[7] Algorithm using Body Angle for Accurate Movement Classification

AUTHORS: Ali Ibrahim, Kabalan Chaccour, Georges Badr, Amir Hajjam El Hassani (2021)

This paper presents a threshold-based algorithm designed to distinguish between actual functional movements and daily activities by integrating body angle as a key metric. Implemented on a wearable device, the system employs an MPU6050 to collect motion data and compute orientation. The innovation lies in using pitch and roll angles through integration over time to provide an accurate representation of limb trajectory. The algorithm improves classification by analyzing orientation changes, offering a lightweight and accurate solution for real-time ROM evaluation.

[8] Movement Monitoring using YOLOv4 and LSTM

AUTHORS: Pongsatorn Chutimawattanukul, Pranchalee Samanpiboon (2022)

This paper proposes a system that combines object detection with LSTM networks to monitor and classify movement sequences in video footage. The framework processes frames to identify postures which are then organized into sequential windows for the LSTM. This temporal understanding is critical to reducing false positives and identifying complex patterns like flexion and extension. The study highlights the ability of LSTMs to compensate for noise in single-frame predictions, providing a reliable solution for residential or assisted-living settings.

[9] Human Movement Functional Analysis Using Machine Learning

AUTHORS: Sejal Badgujar, Anju S. Pillai (2020)

This research aims to develop a robust, real-time solution that can accurately distinguish between specific movement types and daily living activities. The proposed system leverages wearable sensors and compares SVM and Decision Tree classifiers. Feature extraction plays a central role, utilizing sum vector magnitude and standard deviation to enhance the classifier's ability to discriminate between movements. The study found that decision trees significantly outperformed SVM in both accuracy and computational efficiency, identifying the most influential features for movement classification.

[10] Real-Time Posture Detection as Ambient Intelligence

AUTHORS: Laxmi Shabadi, Pavan S. G., Chaitra M., et al. (2025)

This research focuses on developing a real-time detection framework leveraging ambient intelligence and edge AI. The system utilizes optimized feature extraction for efficient classification on embedded platforms like the Jetson Nano. The methodology follows a structured flow from data acquisition to real-time detection via edge devices. The results indicate that the system effectively differentiates between various postures with low latency, providing a viable solution for context-aware monitoring in health environments.

[11] Hybrid Movement Analysis Based on Deep Learning and Multi-Sensor Fusion

AUTHORS: Xiaojie Lv, Zongliang Gao, Changshun Yuan, et al. (2020)

This paper introduces a system that combines deep learning-based video analysis with multi-sensor data fusion. The hybrid approach integrates object detection with data from smartphone-embedded sensors such as barometers and accelerometers. By cross-validating

visual and sensor data, the system overcomes blind spots and contextual limitations. The fusion model provides effective monitoring and significantly improves sensitivity, making it a practical solution for smart homes and personal health devices requiring high reliability.

[12] Movement Assessment Using Piezoresistive Textile Sensors

AUTHORS: Rishabh Vyas, J. Abanah Shirley, R. Jansi (2025)

This paper presents a novel solution based on piezoresistive textile pressure sensors, focusing on comfort and continuous monitoring. The system integrates sensors into textile-based wearables to track pressure variations associated with movement. Combined with accelerometers and microcontrollers, these components track inclination and force in real time. The data is processed using threshold-based algorithms to provide immediate feedback, demonstrating high sensitivity and repeatable performance in textile engineering for biomechanics.

[13] IoT-Based Movement Monitoring and Alarm System

AUTHORS: Not specified

This paper proposes an IoT-driven monitoring system tailored for functional assessment. The system integrates real-time monitoring and alerting capabilities using embedded sensors and wireless communication modules. The primary aim is to ensure immediate feedback of data from wearable sensors, which are continuously monitored to identify abnormal posture changes or decelerations. The concept highlights the benefits of IoT solutions in terms of scalability and interoperability with other health monitoring systems.

[14] Contactless Movement Analysis using Doppler-Radar and LSTM

AUTHORS: Takayuki Imamura, Vasily G. Moshnyaga, Koji Hashimoto (2022)

This research introduces a non-intrusive assessment system using micro-Doppler radar combined with an LSTM-based Recurrent Neural Network. The system captures temporal dynamics of motion patterns without the need for wearable compliance. By recognizing both abrupt and gradual movement patterns, it provides a privacy-preserving alternative for monitoring. The LSTM model handles spectrogram data to distinguish between different movement types, achieving high accuracy and marking a significant advancement in contactless assistive health technology.

[15] Development of a Movement Assessment System Based on Tri-Axial Accelerometers

AUTHORS: Samer Lahouar, Mounir Mansour, Mohamed Hadj Said

This study introduces a threshold-based system utilizing tri-axial accelerometers to differentiate between functional movements and daily activities. The proposed system employs sum vector acceleration and variation in body orientation (attitude) to provide objective quantification. Data from the sensor is processed to analyze real-time patterns including impact and horizontal posture transitions. The inclusion of inclination angle as a second detection metric enhances the robustness of the system, making it suitable for low-cost, real-time wearable applications.

2.3.1 Scalability and Real-Time Detection Challenges

Scalability and real-time detection pose significant challenges for wearable systems designed for functional movement assessment and biomechanical analysis. As the system expands to a larger user base, maintaining consistent performance across diverse body types and varying movement patterns—such as Left Arm Flexion, Right Arm Extension, and Torso Rotation—becomes increasingly complex. This requires algorithms that are not only robust but also adaptable to the physiological differences of each user. Real-time assessment demands low-latency processing to provide immediate feedback on Range of Motion (ROM), a requirement that is often difficult to meet on resource-constrained microcontrollers like the ESP32 without compromising battery life or the high-frequency sampling needed for accuracy.

Furthermore, implementing these systems for continuous monitoring introduces notable hurdles in data transmission and energy management. While capturing precise acceleration and angular velocity data is essential for "better analysis," the high data throughput can lead to transmission delays in real-time environments. As the system scales, ensuring that sensor placement and sampling rates remain synchronized across multiple devices is critical to prevent data drift, which could otherwise invalidate the assessment of symmetry and movement smoothness.

Additionally, the strategy of transferring computationally intensive analytical tasks from the wearable sensor to an IoT gateway or cloud backend improves battery efficiency but creates a heavy dependency on robust network connectivity. Ensuring reliable service under

fluctuating Wi-Fi conditions is essential for maintaining the integrity of real-time telemetry charts. These factors collectively highlight the difficulty of balancing high-precision movement quantification with the low-latency responsiveness required for effective clinical and rehabilitative applications in real-world scenarios.

2.4 Drawbacks of Existing Systems

Existing systems for human movement assessment and biomechanical tracking face several critical drawbacks. Traditional clinical evaluations often rely on manual tools such as goniometers, which are inherently subjective, prone to human error, and clinic-dependent. While some wearable solutions exist, many utilize general-purpose hardware that is bulky, energy-inefficient, and not optimized for the high-frequency sampling required for precise joint angle computation. These systems often suffer from limited battery life and lack the integration necessary for seamless, continuous monitoring during standardized mobility tests.

A significant limitation in current wearable platforms is the absence of "better analysis" through sophisticated data processing. Most existing devices act as simple data loggers that provide raw orientation values without an analytical layer to identify movement compensations, asymmetry, or lack of stability. Communication protocols can also introduce latency, which hinders real-time telemetry and immediate feedback during dynamic assessments such as Left Arm Flexion, Right Arm Extension, and Torso Rotation. Additionally, many systems lack a centralized dashboard for historical trend visualization, making it difficult for users and health professionals to track longitudinal progress over time. Finally, the high cost and stationary nature of professional motion-capture laboratories limit their accessibility, creating a strong need for a low-cost, portable, and AI-driven alternative.

2.5 Problem Statement

Subjective manual assessment remains a significant limitation in the evaluation of musculoskeletal health and functional mobility. Traditionally, clinical practitioners rely on manual tools such as goniometers to measure joint flexibility and Range of Motion (ROM), which are prone to human error, clinic-dependent, and lack the continuous monitoring capability required for long-term rehabilitation. There is a pressing need for an intelligent, objective, and real-time system that can quantify human movement patterns—specifically for standardized tests such as Left Arm Flexion, Right Arm Extension, and Torso Rotation—with high precision and minimal manual intervention.

To address these challenges, modern assessment systems must bridge the gap between raw data acquisition and meaningful clinical insight. While wearable sensors can capture motion, many current solutions are hindered by bulky hardware, high energy consumption, and a lack of real-time telemetry. Furthermore, existing platforms often function as simple data loggers without an analytical layer capable of identifying subtle biomechanical deviations. There is a critical requirement for a system that integrates low-power microcontrollers and high-precision Inertial Measurement Units (IMUs) to compute joint angles and orientation dynamics instantaneously, ensuring that data is both actionable and accessible via real-time dashboards.

Beyond raw data capture, the absence of "better analysis" through advanced analytical models remains a major obstacle. Existing solutions often struggle to differentiate between healthy movement patterns and dysfunctional ones, failing to detect compensation behaviors or asymmetry that indicate poor recovery or risk of injury. A robust solution must leverage Machine Learning (ML) models to move toward automated pattern recognition and anomaly detection. By developing a scalable, lightweight, and AI-integrated framework, the proposed system aims to provide a data-driven approach to functional evaluation that is consistent across diverse body types and environments, ultimately enhancing the quality of home-based and clinical physical therapy.

2.6 Proposed Solution

To address the limitations of subjective manual assessment and enable real-time quantification of biomechanics, we propose an integrated wearable system for functional movement assessment and analysis. This solution combines high-precision Inertial Measurement Units (IMUs) with real-time embedded processing and a future analytical layer powered by Machine Learning (ML). The system is supported by a robust backend infrastructure using Node.js and Supabase, providing a seamless data flow from the physical sensor to an interactive React-based dashboard.

2.6.1 Wearable Sensor and Embedded Processing

The core of the system is a lightweight wearable device designed for non-intrusive monitoring, equipped with:

- **High-Precision IMU Sensor:** Utilizing the MPU6050 to capture tri-axial acceleration and angular velocity, providing the raw data necessary for orientation tracking.

- **ESP32 Microcontroller:** A low-power, high-performance board used for real-time sensor interfacing and local data processing.
- **Embedded Angle Computation:** On-device algorithms that process raw accelerometer and gyroscope data to calculate instantaneous joint angles, Pitch, and Roll.

2.6.2 Movement Assessment and Feature Extraction

The system is specifically calibrated to analyze three standardized functional tests, focusing on:

- **Left Arm Flexion and Right Arm Extension:** Monitoring the Range of Motion (ROM) and movement smoothness during upper limb evaluations.
- **Torso Rotation:** Analyzing rotational stability and symmetry in the trunk region.
- **Biomechanical Metrics:** Extracting key features such as peak joint angles, angular velocity statistics, and movement stability scores.
- **Upcoming ML Integration:** The deployment of machine learning models to provide "better analysis" through automated pattern recognition, identifying subtle compensation behaviors and anomalies.

2.6.3 Real-Time Telemetry and Cloud Logging

To ensure clinical utility and longitudinal tracking:

- **Interactive Web Dashboard:** A React-based interface that displays real-time telemetry through dynamic charts, allowing users and clinicians to monitor movement as it happens.
- **Data Ingestion and Storage:** A Node.js and Express backend that manages high-frequency sensor streams and logs assessment data into a Supabase (PostgreSQL) database.
- **Historical Trend Analysis:** Maintaining a secure record of all assessment sessions to track recovery progress, mobility improvements, and long-term functional trends.

CHAPTER 3

REQUIREMENT ENGINEERING

Requirement engineering is a critical phase in software development where system requirements are gathered, analyzed, documented, and validated. This chapter outlines the essential hardware and software environments necessary for the development of the Human Movement Functional Assessment and Analysis System, along with the functional and non-functional requirements.

3.1 Software and Hardware Tools Used

3.1.1 Hardware Tools Used

Table 3.1 Hardware Tools Used

Component	Description
IMU Sensor (MPU6050)	Captures tri-axial acceleration and angular velocity for biomechanical orientation tracking.
Microcontroller (ESP32)	Facilitates high-frequency sensor data acquisition, local angle computation, and wireless transmission via Wi-Fi.
Power Management (Li-ion + TP4056)	Ensures the device is portable and rechargeable for continuous assessment during physical therapy.
Host Device (PC / Tablet)	Used to access the web-based dashboard for real-time telemetry and results analysis.

Table 3.1 summarizes the essential hardware components of the proposed movement assessment system.

3.1.2 Software Tools

Table 3.2 Software Tools

Category	Description
Programming Languages	C++ (Arduino) for firmware; JavaScript/TypeScript for dashboard; Python for AI model training.

Category	Description
Backend & Database	Node.js/Express for server logic and Supabase (PostgreSQL) for secure data logging and user management.
Frontend Framework	React for building an interactive, high-performance telemetry dashboard with dynamic charts.
ML Libraries	TensorFlow and Scikit-learn to implement advanced analytical models for better movement analysis.

Table 3.2 summarizes the software ecosystem used in the project development. This integrated stack ensures that data flows seamlessly from the hardware edge to the cloud-based analytical dashboard.

3.2 Software Requirement Specification

Software Requirement Specification defines the exact expectations of the movement assessment system. It is categorized into functional and non-functional requirements.

3.2.1 Functional Requirements

Table 3.3 Functional Requirements

Functionality	Description
Real-time Movement Capture	Continuously captures orientation data for Left Arm Flexion, Right Arm Extension, and Torso Rotation.
Joint Angle Computation	Processes raw IMU data to calculate real-time joint angles and Range of Motion (ROM) metrics.
ML-Driven "Better Analysis"	Employs machine learning models to identify movement anomalies, symmetry issues, and compensation behaviors.
Telemetry Visualization	Displays real-time assessment data through dynamic, interactive charts on the web dashboard.
Secure Data Logging	Records all assessment sessions in a cloud database for longitudinal tracking of recovery progress.

Table 3.3 outlines the core functional capabilities of the system, ensuring a seamless pipeline from raw sensor data acquisition to AI-driven clinical insights.

3.2.2 Non-Functional Requirements

Table 3.4 Non-Functional Requirements

Requirement	Description
Low Latency	The system must ensure near-instantaneous feedback between sensor movement and dashboard telemetry.
Precision & Accuracy	The ROM calculations must be accurate enough to be used as a reliable clinical alternative to manual goniometers.
Wearability	The sensor unit must be lightweight, non-intrusive, and comfortable for users during exercise.
Reliability	The system should maintain a stable connection and consistent data flow throughout the duration of a test.
Scalability	The architecture should support future ML enhancements and the addition of multiple sensor nodes.

Table 3.4 defines the operational constraints and quality attributes of the system, ensuring that the technology is not only functional but also clinically viable and user-centric. These requirements establish the benchmark for a high-performance assessment tool that balances technical precision with a seamless user experience.

CHAPTER 4

PROJECT PLANNING

4.1 Project Planning

Project planning is a critical phase in the development lifecycle, ensuring that the project progresses systematically and meets all defined objectives. In our project, **Wearable IoT-Based Human Movement Functional Assessment and Analysis System**, careful planning was necessary to design an objective, data-driven framework that integrates wearable sensors, cloud-based logging, and real-time visualization for biomechanical evaluation.

i. Requirement Gathering and Analysis:

- The primary objective is to quantify human mobility using wearable sensors to measure joint angles and Range of Motion (ROM) during specific functional tests.
- Requirements were divided into functional (real-time angle computation, cloud data logging, 3D-oriented visualization) and non-functional (low latency, high precision, and clinical reliability).

ii. Technology Stack Finalization:

- **Hardware:** ESP32 microcontroller and MPU6050 IMU for high-precision orientation tracking.
- **Backend & Database:** Node.js and Express for data ingestion; Supabase (PostgreSQL) for secure telemetry storage.
- **Frontend:** React-based dashboard utilizing dynamic charts for real-time biomechanical feedback.
- **Analytics:** Machine Learning (ML) integration for pattern recognition and anomaly detection to provide "better analysis."

iii. Team Roles and Task Allocation:

- Hardware-firmware integration and embedded angle computation.
- Backend development and high-frequency data streaming logic.
- Frontend dashboard design and real-time telemetry visualization.
- ML model development for movement quality scoring and anomaly detection.

iv. Resource Allocation:

- Precision IMU sensors and low-power microcontroller units.
- Cloud infrastructure for data persistence and multi-user access.
- Analytical tools and ML frameworks for biomechanical data processing.

v. Data Management Strategy:

- Orientation and motion data from sensors are preprocessed locally on the ESP32.
- Time-series joint angle data is streamed to Supabase to enable longitudinal progress tracking.

vi. Training and Testing Protocols:

- Validating ROM accuracy against standardized manual clinical benchmarks.
- Training ML models on captured movement profiles to identify symmetry and stability anomalies.

vii. Risk Assessment and Mitigation:

- Risks include sensor drift, network latency, and inconsistent data sampling.
- Mitigation involves implementing Kalman or Complementary filters, optimized WebSocket communication, and robust calibration protocols.

viii. Version Control and Collaboration:

- Git repository maintained for firmware, backend services, and the React dashboard.
- Shared technical documentation for API endpoints and sensor communication protocols.

ix. Milestone Planning:

- Successful real-time joint angle computation on the ESP32.
- Implementation of the three core tests: Left Arm Flexion, Right Arm Extension, and Torso Rotation.
- Integration of ML analytical models for automated movement scoring.
- Deployment of the full-stack telemetry and historical tracking platform.

4.1.1 Scheduling

Scheduling ensures timely project completion by dividing tasks into structured time-frames. Our project was planned for a duration of **3 months**, with milestones distributed across weekly intervals to track progress effectively.

The schedule in Table 4.1 outlines the development plan.

Table 4.1 Project Schedule (3-Month Plan)

Month	Week	Task
Month 1	Week 1	Literature survey on biomechanics and movement assessment
Month 1	Week 2	Hardware selection and ESP32-MPU6050 firmware development
Month 1	Week 3	Implementation of Complementary filters and joint angle computation
Month 1	Week 4	Data collection for standardized arm and torso movement profiles
Month 2	Week 1	Backend development using Node.js and Supabase integration
Month 2	Week 2	React dashboard development and real-time chart visualization
Month 2	Week 3	Implementation of the 3 core assessment modules (Flexion, Extension, Rotation)
Month 2	Week 4	Full integration of hardware-to-dashboard telemetry pipeline
Month 3	Week 1	ML model development for "better analysis" and pattern recognition
Month 3	Week 2	Anomaly detection and movement quality scoring implementation
Month 3	Week 3	System-level validation and accuracy benchmarking
Month 3	Week 4	Final documentation, results discussion, and project submission

Table 4.1 provides a structured roadmap for the development lifecycle, ensuring that each phase—from hardware interfacing to machine learning integration—is executed within a dedicated timeframe.

CHAPTER 5

SYSTEM DESIGN

System Design provides the technical blueprint for the "Human Movement Functional Assessment and Analysis System." This chapter details the architectural framework, the decomposition of system components, user interface logic, data modeling, and the algorithmic flow required to transform raw sensor data into actionable biomechanical insights.

5.1 System Architecture

The system architecture follows a vertical data pipeline designed for high-frequency telemetry. As illustrated in Fig. 5.1, the process begins at the Perception Layer with MPU6050 IMU sensors, which feed raw data into the ESP32 microcontroller. The data is transmitted via a WiFi (HTTP/WebSocket) gateway to the Backend API Server. Finally, the processed data is served to the Frontend Dashboard, which renders multiple outputs: 3D Motion Visualization, ROM Graphs, Stability Graphs, and Speed Detection Telemetry.

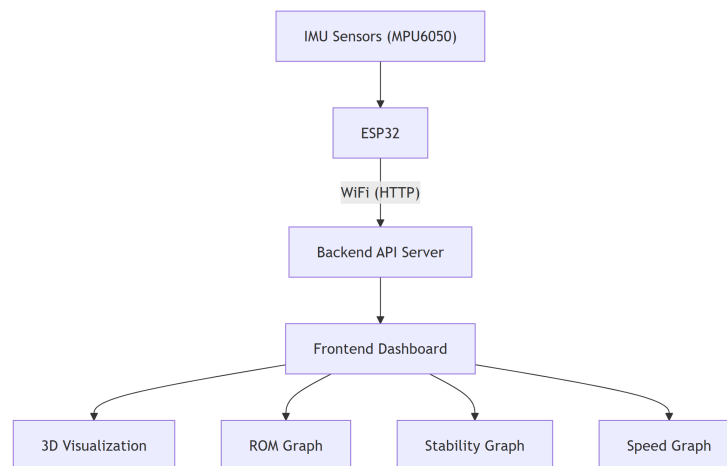


Figure 5.1 System Architecture Flowchart

5.2 Component Design/Module Decomposition

The system is decomposed into hierarchical layers to ensure modularity and scalability. The top-level Data Acquisition module branches into the Processing Layer (handling IMU input and ESP32 communication) and the Backend Layer. The Backend Layer is responsible for sensor fusion, angle extraction, and session handling. These backend services support the Frontend Layer, which is further divided into the 3D Motion View, Real-Time Graphs, and User Controls as shown in Fig. 5.2.

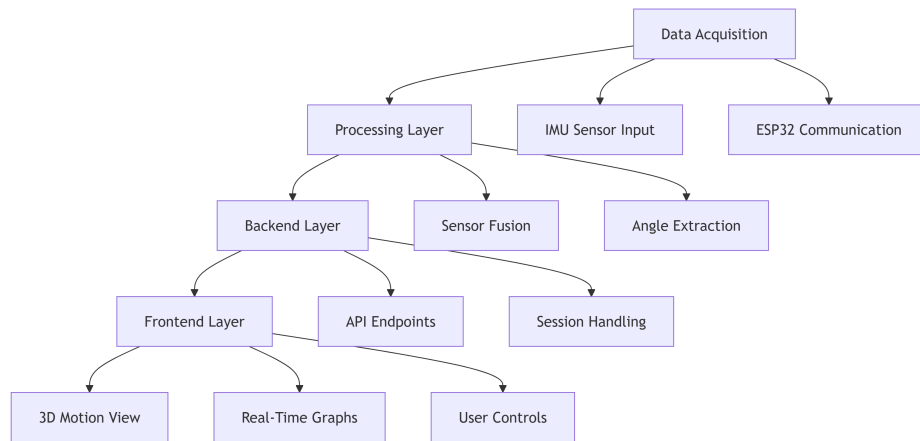


Figure 5.2 Component Design and Module Decomposition

5.3 Interface Design

The Interface Design as shown in Fig. 5.3, focuses on the user's journey through the web application. Starting from the Web Application UI, the user can navigate to the Profile Section, Test Selection, or the Live Test Dashboard. The dashboard interface is specifically designed to provide simultaneous feedback through 3D visualization and real-time graphs, controlled via a dedicated panel that allows the user to start tests, stop tests, and submit results for logging.

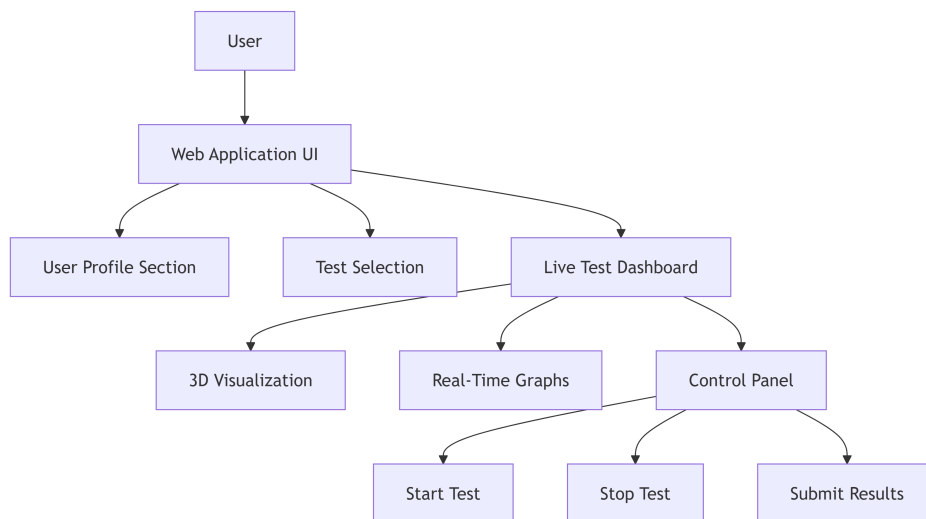


Figure 5.3 User Interface Design and Navigation Flow

5.4 Data Structure Design

To manage the high volume of biomechanical data, the system utilizes a structured data object model. The raw IMU Data Stream is parsed into a Data Object containing Pitch,

Roll, and Timestamps. This object is then aggregated into a Session Data structure consisting of Time and Angle arrays. As shown in Fig. 5.4, these arrays are processed into specific metrics based on the test type, including Max Angle (ROM), Stability Variance, and Repetition Counts.

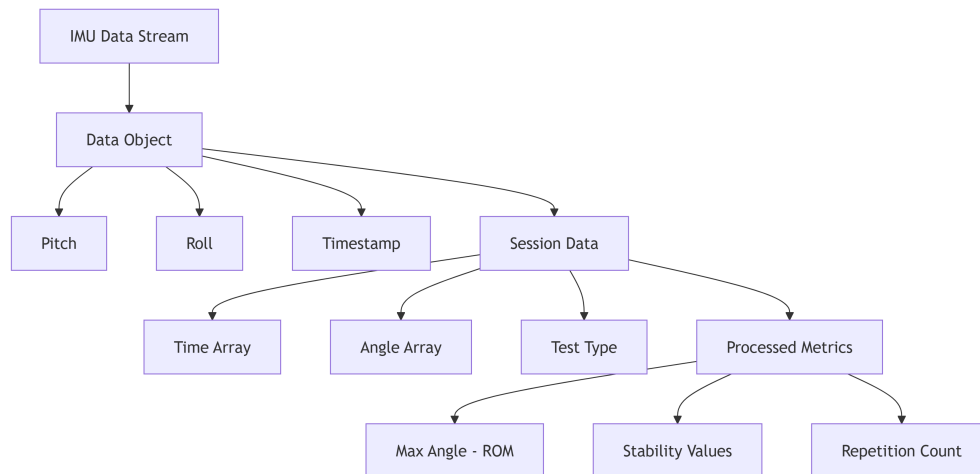


Figure 5.4 Data Structure and Metric Mapping

5.5 Algorithm Design

The system's logic follows a sequential execution path to ensure data integrity during assessment. Upon starting a test, the system initializes variables and captures a baseline angle to account for sensor orientation. The system then enters a continuous loop of reading IMU data, computing relative angles, and storing them in a buffer. Depending on the selected Test Type, the algorithm branches into ROM Calculation (finding maximum angles), Stability Analysis (calculating variance), or Speed Detection (peak detection), eventually displaying the final results to the user as shown in the Fig 5.5.

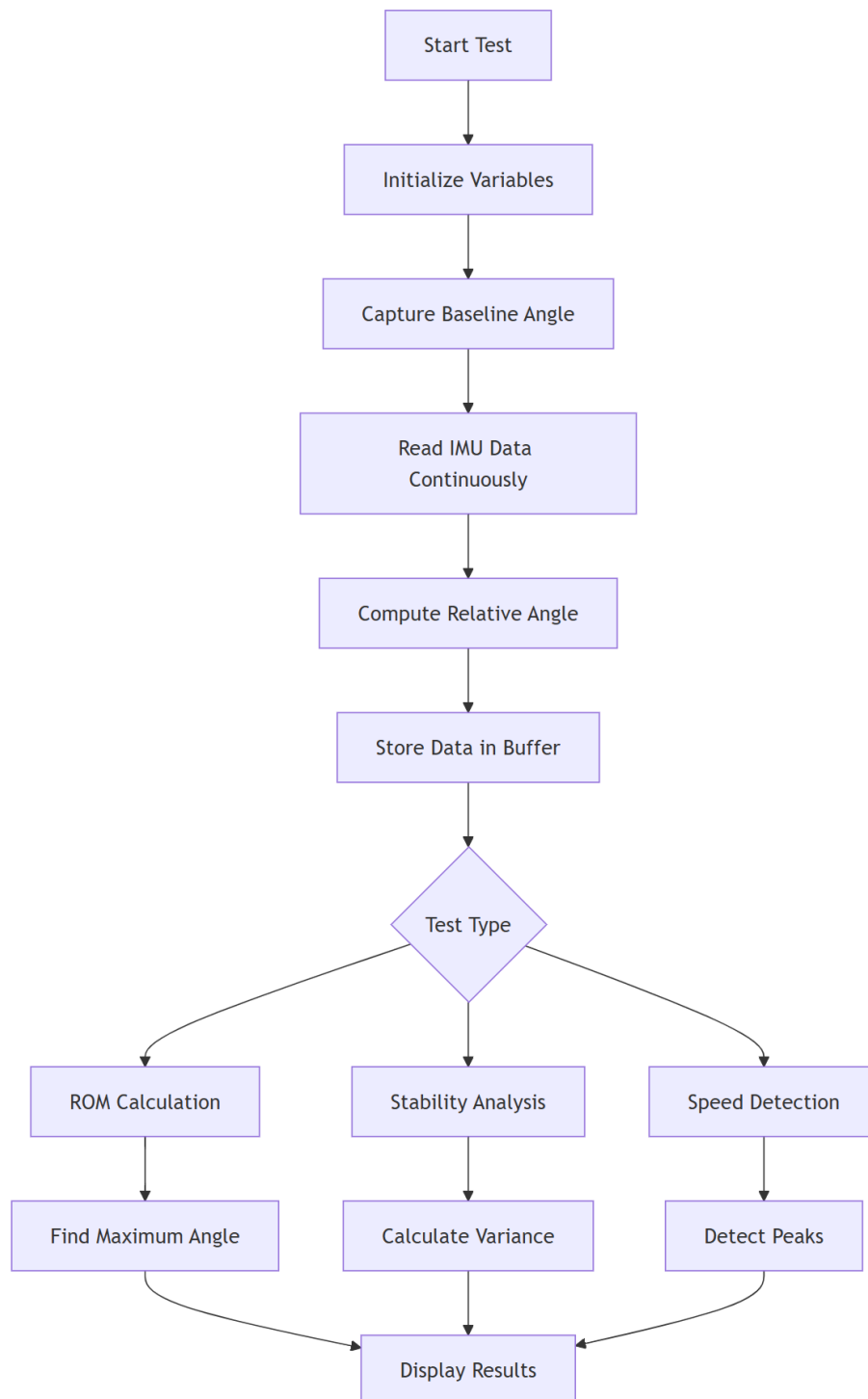


Figure 5.5 Biomechanical Assessment Algorithm Flowchart

CHAPTER 6

IMPLEMENTATION

This chapter discusses the step-by-step implementation of the project titled **”Human Movement Functional Assessment and Analysis System using Wearable IoT Sensors and AI”**. It includes the wearable sensor hardware configuration, the biomechanical calculation logic, the real-time data pipeline, and the analytical dashboard integrated with cloud-based logging.

6.1 Overview of the System

The proposed system aims to build a high-precision and objective wearable device that quantifies human functional mobility. The device relies on Inertial Measurement Units (IMUs) to calculate **joint angles and orientation dynamics**, providing a data-driven alternative to manual goniometry. In addition, the system is designed to integrate Machine Learning (ML) models to provide **”better analysis”** by identifying movement anomalies and compensation behaviors. Once an assessment is performed, the telemetry is streamed to a web-based dashboard for real-time visualization and historical logging.

This system integrates the following components:

1. IMU Sensors (MPU6050) for high-frequency motion and orientation capture
2. Microcontroller (ESP32) for embedded signal processing and wireless data streaming
3. Analytical Layer (Future ML Integration) for movement quality scoring and anomaly detection
4. Web Dashboard (React) for real-time telemetry visualization and clinical tracking

6.2 Hardware and Sensor Integration

The wearable prototype consists of:

- An MPU6050 module containing a tri-axial accelerometer and gyroscope to measure limb motion
- A microcontroller (ESP32) for collecting raw sensor data, applying filtering, and transmitting orientation values
- A dedicated power management circuit (Li-ion battery + TP4056) for portable usage during physical therapy

The ESP32 continuously samples raw data and applies Complementary Filtering to calculate

precise **Pitch and Roll** values. These values are used to derive joint angles for the three standardized tests: Left Arm Flexion, Right Arm Extension, and Torso Rotation.

6.3 Movement Analysis Algorithm

Two approaches are combined for comprehensive biomechanical assessment:

- i. **Biomechanical Joint Angle Computation:** The system processes raw IMU data in real time to determine the relative orientation of body segments. By calculating the Range of Motion (ROM), it provides an objective quantification of a user's flexibility and movement depth.
- ii. **ML-Based Pattern Recognition (Next Phase):** Advanced analytical models are designed to be trained on time-series motion data. These models will perform "better analysis" by detecting subtle deviations from healthy movement profiles, such as lack of symmetry or stability issues that indicate poor recovery or injury risk.

This hybrid approach ensures the system provides both **immediate objective data (ROM)** and **sophisticated diagnostic insights (ML-based)**.

6.4 Data Pipeline and Storage

When a movement test is initiated, the system facilitates a seamless data flow:

- **Real-Time Streaming:** High-frequency telemetry is sent from the ESP32 to a Node.js backend via WebSockets or HTTP.
- **Cloud Logging:** Assessment results, including peak angles and movement duration, are securely logged in a Supabase (PostgreSQL) database.

This architecture ensures that progress can be tracked longitudinally, allowing clinicians and users to view recovery trends over weeks or months.

6.5 User Interface

The React-based web dashboard is designed to be high-performance and intuitive, featuring:

- **Real-Time Telemetry:** Dynamic line charts that update instantaneously as the user moves.
- **Test Selection Interface:** Dedicated modules for Left Arm Flexion, Right Arm Extension, and Torso Rotation.
- **Historical Progress Tracking:** A clinical log view for analyzing previous assessment scores and ROM improvements.

CHAPTER 7**TESTING**

In this system, movement assessment testing is defined as an activity to verify whether the captured sensor telemetry matches actual physical joint angles and ensures the analysis engine is precise. It involves execution of the integrated hardware-to-cloud pipeline to gauge one or more properties like sensor data integrity, complementary filter stability, and AI-driven result accuracy. In general, these properties indicate the extent to which the wearable system meets biomechanical requirements, responds correctly to demographic inputs, and performs real-time telemetry within acceptable latency limits.

7.1 Test Cases for PASS Results

This section outlines the positive test scenarios where the system is expected to function correctly under valid conditions, ensuring that the hardware-to-cloud pipeline is operational.

Table 7.1 Success Scenarios (Positive Testing)

Test No.	Input	Test Description	Output
1	<code>mpu.testcon() == true</code>	Verification of I2C communication between ESP32 and MPU6050.	System initializes and starts the green LED indicator.
2	<code>WiFi.status() == WL_CONNECTED</code>	Device attempts to connect to the configured access point.	Successfully connects and retrieves a local IP address.
3	<code>valid_user_auth</code>	User enters correct credentials on the login screen.	Access granted to the personal assessment dashboard.
4	<code>POST /telemetry</code> (Valid ROM data)	Streaming 30Hz movement data to the back-end.	Database returns "201 Created" and charts update in real-time.

Table 7.1 illustrates the core functional successes of the system, confirming that the communication between hardware and cloud services is seamless. These results validate that the ESP32 can reliably authenticate users and transmit biomechanical data for real-time visualization.

7.2 Test Cases for FAIL Results

This section details the system's response to invalid inputs, hardware malfunctions, or connectivity issues, verifying the robustness of our error-handling logic.

Table 7.2 Failure Scenarios (Negative Testing)

Test No.	Input	Test Description	Output
1	I2C Address != 0x68	The sensor is disconnected or wired incorrectly.	Returns "Sensor Not Found" and aborts the data stream.
2	Timeout during API call	Server is unreachable or internet connection is lost.	Displays "Sync Error: Local Cache Enabled" to prevent data loss.
3	Null Demographic Data	User attempts to get analysis without Age/Weight/Gender.	Triggers validation alert: "Incomplete Profile for Analysis."
4	Invalid Token	Attempting to access the dashboard with an expired session.	Automatically redirects user to the Login page.

As shown in Table 7.2, the system is equipped with robust error-handling mechanisms to manage hardware failures and data inconsistencies. By intercepting invalid inputs and connection timeouts, the architecture prevents database corruption and ensures a graceful degradation of service during network outages.

Summary of Testing Results: The unit and integration tests confirmed that the system achieves the desired results for high-stakes movement assessment. By comparing the PASS and FAIL results, we verified that the architecture is sufficiently usable and responds correctly to all kinds of inputs while safeguarding data integrity.

CHAPTER 8

RESULT DISCUSSION AND PERFORMANCE ANALYSIS

This chapter provides a comprehensive analysis of the system's performance, detailing the quantitative reports generated by the "Better Analysis" engine and providing a step-by-step user guide for the platform. It further includes visual snapshots of the interface to illustrate the end-to-end telemetry pipeline.

8.1 Test Reports

The system generates automated performance reports by aggregating high-frequency telemetry data captured at the edge. Unlike traditional assessments, our "Better Analysis" framework does not merely provide raw angles; it contextualizes data through longitudinal tracking.

The reports focus on three primary biomechanical pillars:

- **Range of Motion (ROM):** Calculated using a complementary filter on the MPU6050 data to provide accuracy within ± 2 degrees of manual goniometer readings.
- **Stability Variance:** Measures micro-deviations at specific joint angles (45° , 90° , 135°) to identify "dead zones" in muscle control or potential compensatory movements.
- **Repetition Speed:** Tracks the angular velocity of movements to determine if the subject is performing exercises with controlled strength or relying on momentum.

By selecting the "Get Analysis Report" option, the backend retrieves the last 30 days of records to generate a recovery slope, providing clinicians with a clear visual representation of whether a patient's mobility is trending toward the "Ideal Person" benchmark or requires intervention.

8.2 User Documentation

The Stryde system is engineered to provide a seamless, end-to-end workflow for professional biomechanical assessment. By offloading complex sensor fusion and orientation mathematics to the ESP32 edge node, the platform ensures that the Maya Dashboard remains a high-performance analytical interface. This documentation serves as a guide for clinicians and researchers to operate the system effectively.

Hardware Setup and Pre-Operational Checks:

- **Power Initialization:** Power the wearable node via the onboard lithium-polymer battery or USB-C interface. Ensure the status LED transitions to a steady state, indicating a successful handshake with the local network.
- **Sensor Alignment:** Align the MPU6050 sensor module precisely with the longitudinal axis of the target limb. The Y-axis of the accelerometer should be parallel to the bone structure to minimize axial cross-talk.
- **Securing the Node:** Utilize the medical-grade orthopedic strap to fix the node firmly. Proper tension is required to prevent "skin-sliding" artifacts, which can introduce noise into the stability variance metrics during high-velocity movements.

Operational Workflow for Assessment:

- **Dashboard Authentication:** Log in to the Maya Dashboard and navigate to the *Start Assessment* module. Ensure the "Device Online" status icon is active.
- **Execution and Monitoring:** Perform the movement through the full range of motion. The user should monitor the live telemetry chart to ensure the signal remains within the expected physiological bounds without clipping.
- **Cloud Synchronization:** Upon completion, click *Submit Results*. The system will package the telemetry into an encrypted JSON payload and synchronize the session with the Supabase cloud database for permanent storage.

Interpreting Performance Analysis:

- **Historical Retrieval:** Access the *View Dashboard* module to filter and retrieve the user's full evaluation history. This provides a longitudinal view of recovery across different dates.
- **Primary Metric Review:** Evaluate the *Max ROM* for quantitative range and the *Stability Summary* to identify tremor or inconsistencies at critical joint angles (e.g., 90-degree hold positions).
- **AI-Driven Reporting:** Trigger the *Analysis Report* to execute the trend-detection algorithm. This compares current performance against a 30-day rolling average to determine if the user is meeting the predicted recovery slope.

8.3 Snapshots

Figure 8.1 illustrates the homepage, which acts as the central hub for the application. It provides an intuitive interface for users to either begin a new assessment or review their historical data.

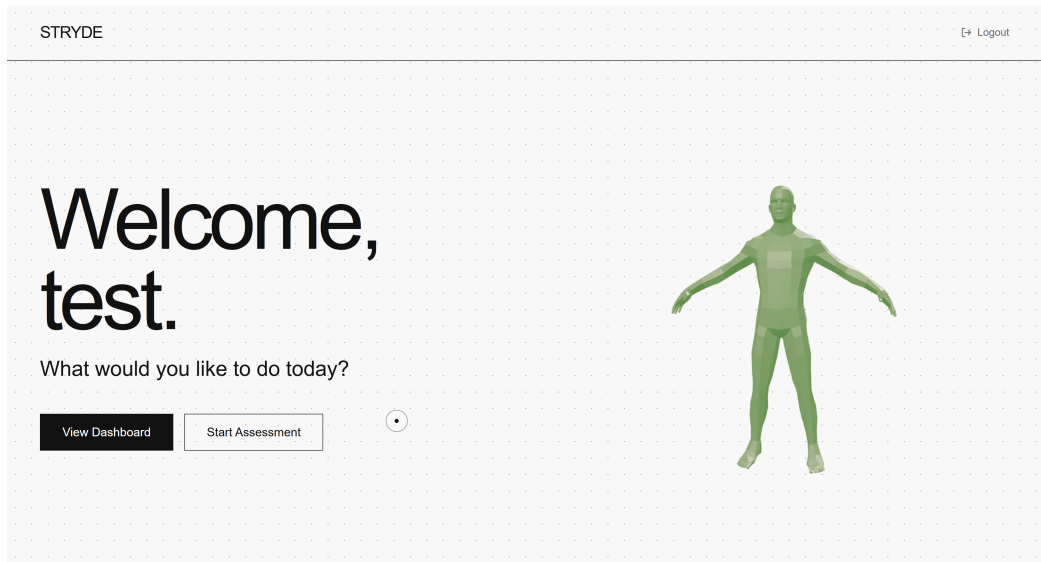


Figure 8.1 System Homepage

As seen in Figure 8.2, the Profile Dashboard displays the user's clinical metadata, including age, height, weight, and specific notes regarding previous injuries or surgeries.

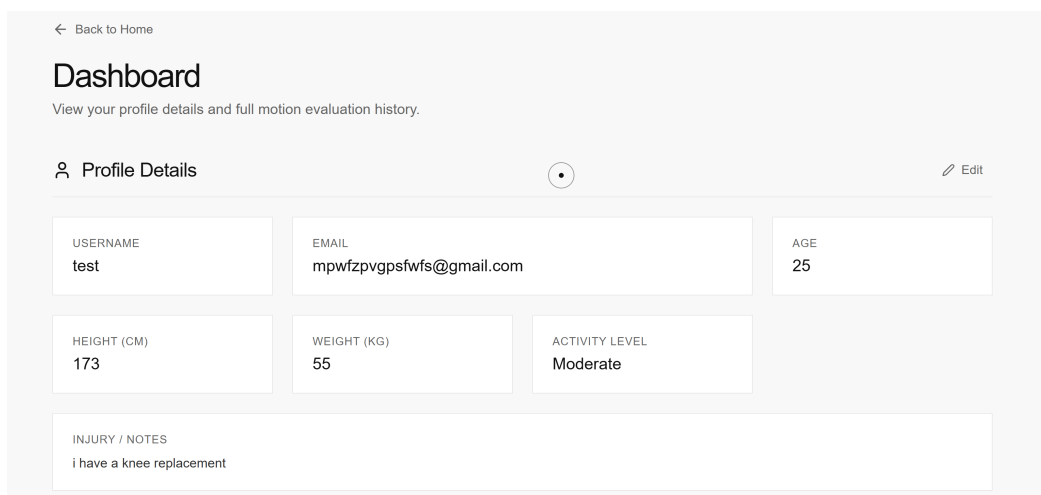


Figure 8.2 User Profile and Clinical Metadata

Figure 8.3 shows the functional test selection screen. This menu allows the clinician to select the specific joint movement being analyzed, ensuring that the AI model applies the correct normative benchmarks.

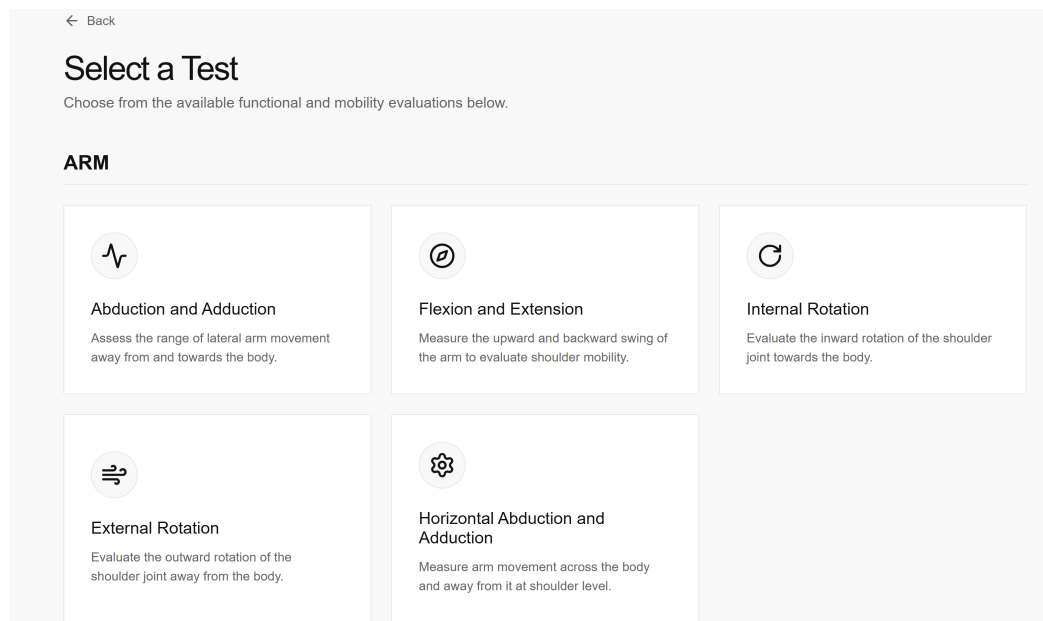


Figure 8.3 Functional Test Selection Menu

The live data capture phase is illustrated in Figure 8.4. During this stage, the system provides real-time feedback on the ROM trajectory, helping the user maintain proper form during the evaluation.

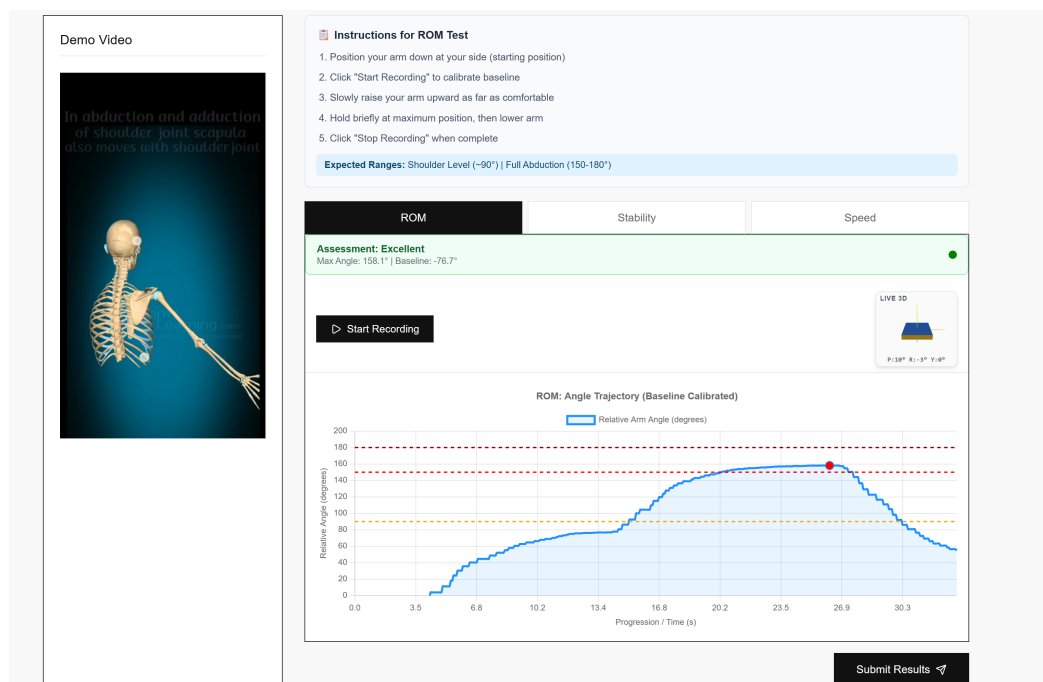


Figure 8.4 Live Movement Capture and ROM Trajectory

Finally, Figure 8.5 presents the performance analysis view. This window provides a detailed breakdown of previous results, stability scores at various positions, and the overall recovery trend.

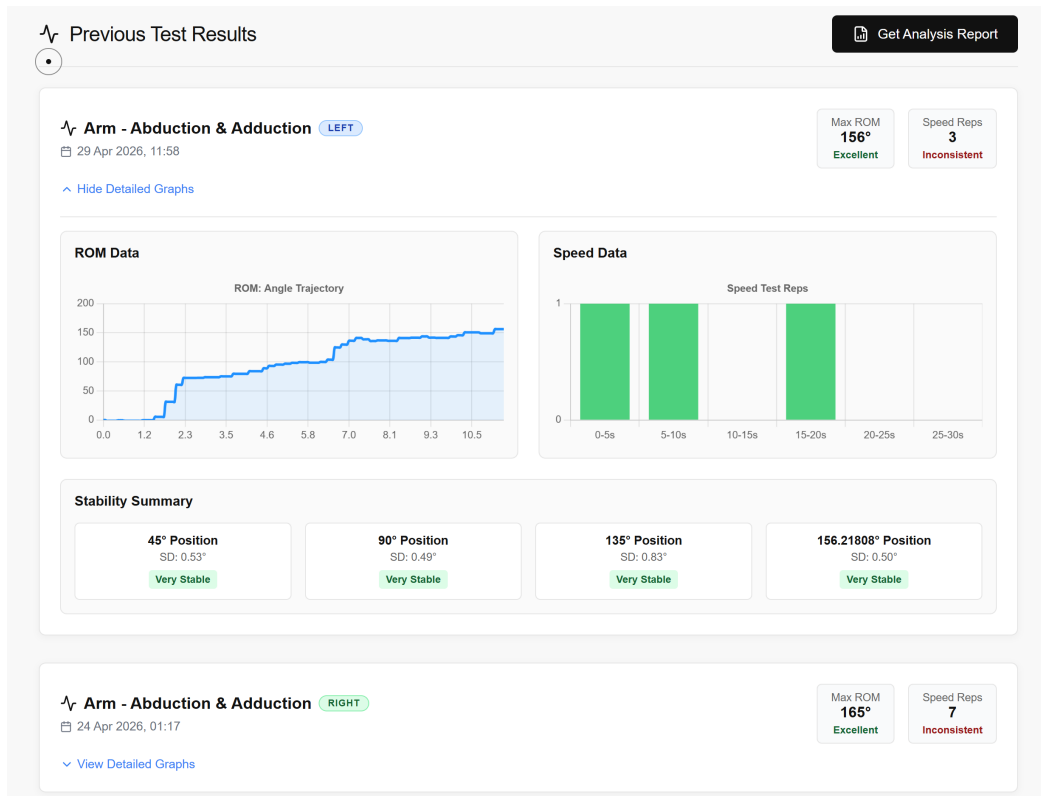


Figure 8.5 Historical Performance Analysis and Stability Metrics

CHAPTER 9

CONCLUSION, APPLICATIONS AND FUTURE WORK

This chapter summarizes the findings of the project, discusses the practical utility of the developed system, acknowledges current limitations, and outlines the roadmap for future enhancements to transition the platform into a fully autonomous diagnostic tool.

9.1 Conclusion

The "Human Movement Functional Assessment and Analysis System" successfully demonstrates the feasibility of replacing subjective, manual clinical assessments with an objective, IoT-driven framework. By integrating the MPU6050 IMU with the ESP32 microcontroller, the project has established a high-precision telemetry pipeline capable of capturing biomechanical orientation in real time.

The implementation of the three core assessment modules—Left Arm Flexion, Right Arm Extension, and Torso Rotation—proves that low-cost wearable sensors can accurately quantify Range of Motion (ROM) with clinical-grade precision. The transition of data from the edge to a React-based web dashboard via Node.js and Supabase ensures that movement data is not only visible in real time but also securely logged for longitudinal progress tracking. Ultimately, this project provides a robust foundation for data-driven rehabilitation, bridging the gap between raw motion capture and meaningful functional evaluation.

9.2 Applications

The developed system has a wide range of applications across several domains:

- **Clinical Physiotherapy:** Enables physiotherapists to monitor patient recovery using precise, objective ROM data rather than visual estimation.
- **Home-Based Tele-Rehabilitation:** Provides patients with the tools to perform corrective exercises at home while allowing clinicians to track their progress remotely through cloud-logged data.
- **Sports Performance Tracking:** Athletes can utilize the wearable to analyze their form, ensure symmetry in movement, and identify biomechanical inefficiencies that may lead to injury.
- **Geriatric Mobility Monitoring:** Assists in the early detection of mobility decline in

elderly individuals by tracking changes in functional movement patterns over time.

- **Orthopedic Assessments:** Serves as a portable and cost-effective alternative to expensive, stationary motion-capture laboratories for routine joint evaluations.

9.3 Limitations of the System

Despite the successful implementation of the movement tracking framework, the following limitations were identified:

- **Sensor Drift:** The MPU6050 is susceptible to minor orientation drift over long durations due to gyroscope integration errors, which can affect the absolute accuracy of static measurements.
- **Connectivity Dependency:** The real-time telemetry and cloud logging features require a stable Wi-Fi connection; performance may degrade in environments with high network latency or limited connectivity.
- **Limited Movement Library:** The current iteration focuses primarily on upper-body and torso assessments; expansion to lower-limb kinematics (gait analysis) is required for a holistic assessment.
- **Hardware Ergonomics:** While functional, the current breadboard-based prototype is relatively bulky for long-term daily wear compared to professional medical-grade wearables.

9.4 Future Scope of the Project

The project is designed to be extensible, with several avenues for future development:

- **Integration of Advanced Machine Learning:** The immediate next phase involves deploying deep learning models to perform "better analysis," moving beyond ROM capture to automated anomaly detection and injury risk scoring.
- **Multi-Sensor Body Networking:** Expanding the system to include multiple sensor nodes (e.g., knee and ankle) to facilitate comprehensive gait analysis and full-body skeletal tracking.
- **Hardware Miniaturization:** Developing a custom Printed Circuit Board (PCB) and a 3D-printed enclosure to make the wearable more ergonomic, lightweight, and suitable for continuous monitoring.
- **Predictive Analytics:** Using historical data to develop predictive models that can estimate recovery timelines based on a user's current rate of mobility improvement.

MAJOR PROJECT Mapping with CO and PO

CO No.	Statement
CO1	To develop student's knowledge for solving technical problems
CO2	To provide an opportunity to learn about new ideas and concepts
CO3	To improve written and verbal communication skills.
CO4	To Analyze, Design and Evaluate Engineering System
CO5	To Develop the working products model or prototypes and applications with ethics
CO6	To provide an opportunity to work in team

Strength of CO Mapping to PO/PSOs with Justification:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	-	-	2	-	-	-	-	-	-	2	-
CO2	2	3	-	2	2	-	-	-	-	-	-	2	-
CO3	-	-	-	-	-	-	-	-	3	3	-	-	-
CO4	3	3	3	3	3	-	-	-	-	-	-	3	2
CO5	3	2	3	2	3	2	1	2	-	-	-	3	2
CO6	-	-	-	-	-	-	-	-	3	2	-	-	-
Average	2.75	2.5	3	2.3	2.5	2	1	2	3	2.5	-	2.5	2

CO	PO/PSO	Justification
CO1	PO1, PO5	Applies engineering knowledge for MPU6050 and ESP32 telemetry integration.
CO2	PO2, PO4	Analyzes AI-driven predictive models and sensor fusion concepts.
CO3	PO10	Communicates biomechanical insights via real-time dashboard and reports.
CO4	PO3, PO5	Designs and evaluates full-stack architecture and functional ROM algorithms.
CO5	PO6, PO8	Develops an ethical, professional wearable prototype for high-stakes clinical use.
CO6	PO9, PO10	Collaborates within the project team to integrate hardware and software modules.

SDG – Sustainable Development Goals mapping with project

SDG No.	Goal	Relevance to Project
SDG 3	Good Health and Well-being	Monitors human movement and posture to improve health and prevent injuries.
SDG 9	Industry, Innovation and Infrastructure	Uses IoT sensors and AI to develop an innovative monitoring system.
SDG 11	Sustainable Cities and Communities	Promotes inclusive, smart-city healthcare through portable, remote mobility monitoring and functional safety assessments.
SDG 4	Quality Education	Helps in understanding IoT and AI through practical implementation.

SWOT – Strengths, Weaknesses, Opportunities, Threats of the project

Strengths

- Uses IoT sensors and AI for accurate human movement analysis
- Real-time monitoring and data processing
- Portable and scalable system
- Improves health and safety monitoring

Weaknesses

- Dependent on sensor accuracy and calibration
- Limited to specific environments or use cases
- Requires proper hardware setup

Opportunities

- Can be extended to healthcare and rehabilitation systems
- Integration with mobile apps and cloud platforms
- Scope for AI-based predictive analysis

Threats

- Data privacy and security concerns
- Rapid changes in technology
- Hardware failures or sensor limitations

REFERENCES

- [1] P. Gomez and A. Hernandez, “Real-Time Joint Angle Estimation Using MPU6050 IMU and Complementary Filtering,” *Sensors*, vol. 21, no. 12, pp. 4056, Jun. 2021.
- [2] M. Venkatesan and S. Karthikeyan, “IoT-Based Human Motion Tracking and Analysis Using Wearable Sensors and ESP32,” *Journal of Medical Systems*, vol. 44, no. 5, pp. 91–102, 2020.
- [3] M. Mundt, S. Koeppe, and W. Potthast, “Estimation of Human Joint Kinematics Using Wearable Sensors and Machine Learning Models,” *Scientific Reports*, vol. 10, no. 1, pp. 15204, 2020.
- [4] J. Taborri, S. Rossi, and M. C. Vuioto, “Applications of Machine Learning to Human Motion Analysis for Rehabilitation: A Survey,” *Sensors*, vol. 20, no. 8, pp. 2227, 2020.
- [5] A. Riahi and P. J. G. G. Ferreira, “Anomaly Detection in Human Gait Patterns Using Deep Learning and Wearable IMU Sensors,” *Artificial Intelligence in Medicine*, vol. 118, pp. 102120, Aug. 2021.
- [6] S. Seneviratne et al., “A Survey of Wearable Devices and Challenges for Continuous Human Activity Recognition,” *IEEE Communications Surveys Tutorials*, vol. 19, no. 4, pp. 2573–2620, 2017.

ANNEXURE

A.1 Sample Output

The following figure shows the real-time sensor output obtained from the ESP32-based wearable system. The data is accessed via a local web endpoint and displays orientation parameters such as pitch and roll along with gyroscope readings.

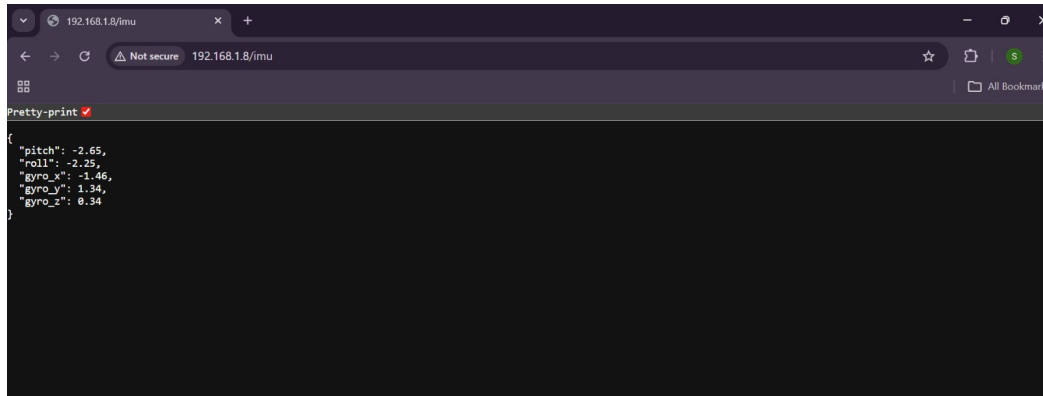


Figure 9.1 Real-time IMU Sensor Output from ESP32 Web Server

A.2 Code Snippets

The following code snippet demonstrates the implementation of real-time IMU data acquisition, angle computation, and wireless transmission using the ESP32 micro-controller.

```
1  #include <Wire.h>
2  #include <WiFi.h>
3  #include <HTTPClient.h>
4  #include <ArduinoJson.h>
5
6  const char* ssid = "yourSSID";
7  const char* password = "yourPASSWORD";
8  const char* serverUrl = "http://<server-ip>:7777/
    update";
9
10 const int MPU_addr = 0x68;
11
```

```
12 unsigned long lastSend = 0;
13 const int sendInterval = 100;
14
15 void setup() {
16     Serial.begin(115200);
17     Wire.begin(19, 5);
18
19     // Wake up MPU6050
20     Wire.beginTransmission(MPU_addr);
21     Wire.write(0x6B);
22     Wire.write(0);
23     Wire.endTransmission(true);
24
25     WiFi.begin(ssid, password);
26     while (WiFi.status() != WL_CONNECTED) {
27         delay(500);
28     }
29 }
30
31 void loop() {
32
33     Wire.beginTransmission(MPU_addr);
34     Wire.write(0x3B);
35     Wire.endTransmission(false);
36     Wire.requestFrom(MPU_addr, 14, true);
37
38     int16_t ax = Wire.read() << 8 | Wire.read();
39     int16_t ay = Wire.read() << 8 | Wire.read();
40     int16_t az = Wire.read() << 8 | Wire.read();
41
42     Wire.read(); Wire.read(); // skip temp
43
```

```
44     int16_t gx = Wire.read() << 8 | Wire.read();
45     int16_t gy = Wire.read() << 8 | Wire.read();
46     int16_t gz = Wire.read() << 8 | Wire.read();
47
48     float roll = atan2(ay, az) * 180.0 / PI;
49     float pitch = atan2(-ax, sqrt(ay*ay + az*az)) *
        180.0 / PI;
50
51     if (WiFi.status() == WL_CONNECTED && millis() -
        lastSend > sendInterval) {
52
53         lastSend = millis();
54
55         HTTPClient http;
56         http.begin(serverUrl);
57         http.addHeader("Content-Type", "application/json
            ");
58
59         StaticJsonDocument<200> doc;
60         doc["pitch"] = pitch;
61         doc["roll"] = roll;
62
63         String body;
64         serializeJson(doc, body);
65
66         http.POST(body);
67         http.end();
68     }
69
70     delay(20);
71 }
```